# Practice Set 6 Paper 4 – Recent Advances

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*Practice paper compiled from the question bank.*

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# Q1. Present concept in management of hypospadias

**Type:** LAQ (25 marks) (embryology and proximal repair also in 2017-Winter Paper 3, Q1; Snodgrass repair in 2023-Winter Paper 3, Q3c)

*Read as "present (current) concepts in the management of hypospadias".*

## Introduction

- Hypospadias is an arrest of the androgen-dependent "closing zipper" of urethral fold fusion (weeks 8–16). Triad: ventral ectopic meatus, ventral curvature, dorsal hooded prepuce.
- 1 in 150–300 male births; distal 70–85%, proximal 10–25%.
- **Goals:** a straight erection, a urethra to the tip (standing voiding, semen delivery) and acceptable appearance, with the fewest operations and a result lasting into adulthood.

## Concept 1: severity is judged in theatre, not by the meatus

- A "distal" meatus may hide absent spongiosum and severe curvature, so classify **after degloving**.

| GMS (glans–meatus–shaft) score | Glans | Meatus | Shaft curvature |
|---|---|---|---|
| 1 | Good size, deep groove | Glanular | None |
| 2 | Adequate, grooved | Coronal | <30° |
| 3 | Small; narrow or flat plate | Mid or distal shaft | 30–60° |
| 4 | Very small; flat plate | Proximal shaft, penoscrotal | >60° |

- Total 3–12; a higher score predicts more complications.
- **POST (plate objective scoring tool):** a low ratio of neomeatus length to glans fusion length means a poor plate, which favours an inlay graft over plain TIP (tubularised incised plate).

## Concept 2: evaluation

- **DSD (disorder of sex development) work-up** if hypospadias plus undescended testis (especially non-palpable), perineal meatus or ambiguity: karyotype, ultrasound for Müllerian remnants, testosterone, LH (luteinising hormone), FSH (follicle-stimulating hormone), AMH (anti-Müllerian hormone).
- No routine VCUG (voiding cystourethrogram) or renal ultrasound. Associations: hernia or hydrocele 9–16%, cryptorchidism about 7% (10% in proximal), enlarged utricle.
- **Aetiology:** causes converge on deficient androgen action (placental insufficiency, assisted reproduction, genes, endocrine disruptors).

## Concept 3: timing and preparation

- Repair at **6–18 months** (AAP (American Academy of Pediatrics): 6–12 months), before genital awareness at about 18 months. One short anaesthetic appears low-risk.
- A straight penis that voids standing may gain little from surgery: counsel honestly.
- **Preoperative androgen** if glans width <14–15 mm: testosterone 2 mg/kg IM (intramuscular) at 5 and 2 weeks before surgery (about 4 mm glans gain). Evidence mixed; two RCTs (randomised controlled trials) showed fewer complications.

## Concept 4: curvature decides the operation

```
Deglove + excise dysplastic dartos + artificial erection
  ├─ straight       → urethroplasty
  ├─ residual <30°  → dorsal midline plication (12 o'clock nerve-free zone)
  └─ >30°           → divide plate → still >30° → ventral corporotomy ± graft → staged repair
```

- Recurrence 28% after plication versus 9% after grafting. Under-correction causes repeated surgery, so correct fully first time.

## Concept 5: choice of technique

| Situation | Current choice |
|---|---|
| Glanular meatus, no curvature | Advancement: MAGPI (meatal advancement and glanuloplasty), MIV (M inverted V) glansplasty; selectively |
| Distal or midshaft, good plate | **TIP (Snodgrass)**: >90% of distal, about 80% of midshaft repairs; Thiersch–Duplay; GAP (glans approximation procedure) for a deep groove |
| Narrow or shallow plate | TIP with **dorsal inlay graft**; onlay island flap; Mathieu (SLAM modification) |
| Proximal, curvature >30° | **Two-stage**: STAG (staged tubularised autograft) or Bracka graft; Byars flaps; distal-first (Ulaanbaatar) |
| Proximal, curvature corrected, good tissue | One-stage onlay or TPIF (transverse preputial island flap); proximal TIP abandoned |

- **Why stage:** a urethroplasty laid over a fresh corporal graft fails more often; staging gives a supple, vascular bed. Stage 2 after ≥6 months.

## Concept 6: protect blood supply

- Magnification, tension-free closure, offset suture lines, **vascularised barrier layer** (dartos, tunica vaginalis), rapidly absorbed fine sutures, no hair-bearing skin, haemostasis.
- Postoperative: stent 3–21 days by complexity, compressive dressing, oxybutynin; antibiotic prophylaxis debated.

## Concept 7: standardised complications and outcomes

- Definitions: glans dehiscence (fusion <2 mm); meatal stenosis (symptoms plus <8 Fr before puberty, <12 Fr after); recurrent curvature >30°.
- TIP meta-analysis (Pfistermuller 2015): complications 10.6%, fistula 5.7%. Proximal complications 23–68%; STAG about 23% (lowest); Byars 43–81%.
- A long, rigid, spongiosum-free neourethra raises resistance (Poiseuille's law), causing fistula and diverticulum. Fistula: see Practice Set 6 Paper 4, Q3d.
- **Also know:** BXO (balanitis xerotica obliterans), excise and stage with buccal graft; redo repairs use buccal (lip inside glans) mucosa, Cecil scrotal parasitisation, tissue expansion.

## Concept 8: follow-up and patient-reported outcomes

- About three-quarters of complications appear by 3 years; stricture and curvature appear late. Follow proximal cases **through puberty**.
- Tools: HOSE (Hypospadias Objective Scoring Evaluation, surgeon), PPS (Penile Perception Score), HOPE (Hypospadias Objective Penile Evaluation, photo panels). Decision aids reduce parental regret.

## Recent advances

- Registries, AI (artificial intelligence) video review, genomic risk profiling, a quality-of-life measure, engineered tissue.

**Diagram to draw:** curvature algorithm and two-stage graft repair.

**Key points to remember:**

- Classify after degloving (GMS); curvature, not meatus, decides one versus two stages.
- TIP for distal; staged graft repair (STAG/Bracka) for severe proximal.
- Hypospadias plus undescended testis: DSD work-up.
- Follow through puberty; use patient-reported outcomes.

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## Q2. Discuss recent advances in tissue engineering with respect to urology

**Type:** LAQ (25 marks) (also answered in 2023-Winter Paper 1, Q1 and 2022-Summer Paper 4, Q2; this answer stresses recent advances)

### Definition and rationale

- **Tissue engineering** applies engineering and life sciences to build biological substitutes that restore, maintain or improve tissue function (Langer and Vacanti, 1993).
- **Why in urology:** bowel in the urinary tract causes mucus, stones, metabolic acidosis, infection and cancer; grafts for long urethral defects are scarce. Aim: replace "wrong tissue" with "right tissue".

### Basic principle (the triad)

```
Biopsy → cells expanded in culture → seeded on scaffold (+ signals)
       → conditioned in bioreactor → implanted → host vascularises and remodels
```

- **Unseeded** matrices depend on host cells migrating in, so they suit only short onlay defects (about 0.5 cm for tubes in animal work). Longer or tubular gaps need **seeded** scaffolds, or they fibrose and contract.

### Recent advances

#### 1. Cell sources

| Source | Advance and limitation |
|---|---|
| Autologous urothelial and smooth muscle cells | Standard; unusable if the bladder is malignant |
| Adult stem cells: bone marrow or adipose mesenchymal, muscle-derived, **urine-derived** | Easy, non-invasive harvest; act mainly by paracrine growth factors |
| **iPSCs (induced pluripotent stem cells)** | Adult cells reprogrammed; patient-specific, no ethical issue; tumour risk |
| Embryonic stem cells | Pluripotent; ethics, teratoma risk |

- Gene-edited and gene-modified cells are being explored. Cell sheet engineering builds layers without a scaffold.

#### 2. Scaffolds

- Requirements: biocompatible, degrade as tissue forms, porous for cells and vessels, mechanically suited (compliant bladder, non-collapsing tube).

| Type | Examples |
|---|---|
| Acellular matrices | Bladder acellular matrix, SIS (small intestinal submucosa), decellularised dermis: keep native architecture and growth factors |
| Synthetic polymers | PGA (polyglycolic acid), PLA (polylactic acid), PLGA (copolymer): controlled shape and degradation |
| **Newer** | Electrospun nanofibres and hydrogels mimicking extracellular matrix; **"smart" scaffolds** releasing growth factors such as VEGF (vascular endothelial growth factor); silk fibroin |

#### 3. Bioreactors

- Perfuse nutrients through thick constructs and apply **cyclic stretch and filling**, so engineered bladder muscle is conditioned before implantation.

**4. Solving vascularity (the main barrier)**

- Oxygen diffuses only about 100–200 µm, so thick constructs die centrally.
- Strategies: omental wrap (Atala), angiogenic factors, co-seeding endothelial cells, printed vascular channels.

**5. 3D bioprinting**

- Cells in hydrogel **"bio-ink"** laid layer by layer from CT (computed tomography) or MRI (magnetic resonance imaging) data, placing several cell types precisely with built-in channels.
- Bioprinted urethra and bladder constructs are at the animal stage. Printed non-living kidney and prostate models already aid planning and teaching.

**6. Organoids**

- Self-organising 3D mini-organs: **kidney organoids** from iPSCs model disease and drug toxicity; **bladder and prostate tumour organoids** allow personalised drug testing.

**7. Whole-organ decellularisation**

- Kidneys are stripped of cells by detergent perfusion, leaving the matrix and vascular tree, then recellularised with the patient's cells; still at the animal stage. Renal assist devices are a parallel route.

## Clinical applications: where we stand

| Organ | Status |
|---|---|
| **Bladder** | **Atala (2006):** seeded collagen–PGA construct with omental wrap in **7 myelomeningocele patients**, compliance improved. Industry phase II trial: **no significant gain in compliance or capacity**. Still experimental (Chapter 48) |
| **Conduit** | Neo-Urinary Conduit trial (NCT01087687): survived about 250 days on average; **half developed stenosis or stricture**; stiff material, no blood supply (Chapter 135) |
| **Urethra** | Acellular matrix onlay for short strictures; **Raya-Rivera (2011):** seeded tubular PGA–PLGA urethras in **5 boys**, functional up to 6 years; tissue-engineered oral mucosa avoids donor-site morbidity |
| **Sphincter, detrusor** | Autologous muscle-derived cells for SUI (stress urinary incontinence); for underactive detrusor, open-label studies improved voiding efficiency (Chapter 111) |
| **Vagina** | Engineered vaginas in MRKH (Mayer–Rokitansky–Küster–Hauser) syndrome (Raya-Rivera, 2014); off-the-shelf SIS grafts (Chapter 59) |
| **Penis, testis** | Corporal tissue (animal); stem cells for ED (erectile dysfunction) investigational; spermatogonial stem cell banking |

- An engineered bladder gives **storage only**: innervation for voluntary emptying cannot yet be recreated (Chapter 111).
- **Also know:** robotic retrieval of a **vascularised composite bladder allograft** (Chapter 135) is a transplant, not engineered, route to "right tissue", needing immunosuppression.

## Challenges

- Vascularity, innervation, contractility; fibrosis and contracture.
- Scale-up, cost, culture time, regulation; stem cell tumour risk.
- Promising pilots followed by disappointing controlled trials.

## Future

- Vascularised bioprinted tissue, off-the-shelf cell-free smart scaffolds, combined gene and cell therapy.

**Diagram to draw:** biopsy → culture → seeding on scaffold → bioreactor → implant with omental wrap.

> **Key points to remember:**
> - Triad: cells, scaffold, signals; vascularity (100–200 µm diffusion) is the main barrier.
> - Atala 2006 promising; phase II trial no gain; conduit trial half stenosed.
> - Raya-Rivera 2011: seeded urethras in 5 boys.
> - New: iPSCs, urine-derived cells, bioprinting, organoids.

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## Q3a. Newer methods of stone analysis and their significance in management of calculus disease.

**Type:** SAQ (10 marks) (also answered in 2024-Winter Paper 4, Q3a)

### Why analyse stones

- AUA (American Urological Association) and EAU (European Association of Urology): analyse a stone at least once in every former; **repeat if stones recur on treatment** (evolving metabolic condition).
- Composition points to cause: **uric acid** suggests metabolic disease, **cystine** a genetic one, **struvite** infection. Cystine, struvite, 2,8-DHA (2,8-dihydroxyadenine) and xanthine are diagnostic in themselves.

### Why wet chemical analysis is obsolete

- Detects ions, not **crystal phases**: cannot separate COM (calcium oxalate monohydrate) from COD (dihydrate), or apatite from brushite; misses rare stones.

### Newer methods

| Method | Principle | Comment |
|---|---|---|
| **FTIR (Fourier-transform infrared) spectroscopy** | Each compound absorbs a characteristic infrared spectrum | Commonest reference method; tiny sample; detects proteins, drugs |
| **XRD (X-ray diffraction)** | Crystal lattice diffracts X-rays in a unique pattern | Gold standard for crystals; misses amorphous material |
| Polarising microscopy | Crystal birefringence | Cheap; needs experience |
| **Morpho-constitutional analysis (Daudon)** | Stereomicroscopy of surface and section, plus FTIR | Subtypes point to cause, e.g. Ic suggests primary hyperoxaluria |
| SEM (scanning electron microscopy) ± EDX (energy-dispersive X-ray) | Surface and elements | Research |
| Micro-CT (micro-computed tomography), Raman spectroscopy | 3-D structure; scattered light | Research |

### In vivo prediction

- **DECT (dual-energy computed tomography):** two energies (e.g. 80 and 140 kVp) separate **uric acid from non–uric acid** stones with near-perfect accuracy.
- **Hounsfield units:** SWL (shock wave lithotripsy) works best below about 900–1000 HU; cystine and brushite are dense.
- Dusting loses the specimen: collect fragments.

### Significance in management

- **Uric acid:** oral dissolution by alkalinisation (potassium citrate).
- **Struvite:** complete clearance, culture-directed antibiotics.
- **Cystine:** hydration, alkalinisation, tiopronin; avoid SWL.
- **COM, brushite:** hard, SWL-resistant. Brushite can convert from calcium oxalate after SWL or infection.
- **2,8-DHA:** APRT (adenine phosphoribosyltransferase) deficiency; allopurinol or febuxostat. **Drug stones** (indinavir, triamterene): stop the drug.

**Diagram to draw:** FTIR spectrum outline; COM (dumbbell) and COD (envelope) crystals.

**Key points to remember:**

- FTIR and XRD are reference methods; wet chemistry is obsolete.
- DECT separates uric acid stones in vivo.
- Composition guides dissolution, SWL choice and prevention.

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## Q3b. Urolift.

**Type:** SAQ (10 marks) (also answered in 2024-Winter Paper 4, Q3d)

*Read as PUL (prostatic urethral lift, UroLift) for BPH (benign prostatic hyperplasia).*

### Introduction

- A **non-ablative MIST (minimally invasive surgical therapy)**: permanent implants retract the lateral lobes and open the prostatic urethra.
- **FDA (Food and Drug Administration) approval 2013**, extended to ≤100 mL in 2020.
- **Main advantage:** preserves erectile and **ejaculatory** function.

### Implant and mechanism

- Spring-loaded implant with T-shaped ends: **nitinol capsular tab**, **stainless steel urethral end** (resists encrustation), polyethylene terephthalate suture under tension.
- Delivered through a **19-gauge needle** from a cystoscopic device.
- The gland is more compliant than the capsule, so tension compresses the lobe outward; later local atrophy and fibrosis. Implants are MRI (magnetic resonance imaging)-conditional (≤3 T).

### Indications

- **AUA (American Urological Association): 30–80 mL, no obstructing median lobe** (confirm by cystoscopy or ultrasound). EAU (European Association of Urology): <70 mL, no middle lobe.
- Men wanting ejaculation preserved. **Avoid:** median lobe (MedLift data non-randomised), large gland, infection.

### Technique

1. Local anaesthesia; rigid cystoscope.
2. Compress lateral lobe, fire needle, deploy implant; repeat on the other side.
3. **Usually 4–6 implants**, anterolateral, bladder neck to apex.
4. Keep implants **>1.5 cm distal to the bladder neck** (encrustation).
5. Most go home without a catheter.

### Evidence

| | L.I.F.T. (2013) | BPH6 (2015) |
|---|---|---|
| Design | PUL vs sham, 2:1, 19 centres | PUL vs TURP (transurethral resection of prostate), 10 European centres |
| Result | IPSS (International Prostate Symptom Score) about halved, twice the sham effect; Qmax (maximum flow) +4 mL/s | Composite end point 52.3% vs 20.0%; antegrade ejaculation 100% vs 60.6% |
| Retreatment | 13.6% at 5 years | 13.6% vs 5.7% at 2 years |

- TURP wins on IPSS, Qmax and PVR (post-void residual).

### Complications

- Dysuria, haematuria, urgency (<2 weeks); encrustation; rare pelvic haematoma; device misfire.
- Lower major adverse events but **higher retreatment than TURP**.

**Diagram to draw:** axial prostate with implants pulling lateral lobes to the capsule.

**Key points to remember:**

- 30–80 mL, no median lobe; local anaesthesia.
- Ejaculation preserved; retreatment 13.6% at 5 years.
- TURP better on IPSS, flow and durability.

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# Q3c. Combination Intravesical therapy

**Type:** SAQ (10 marks) (also answered in 2025-Winter Paper 4, Q3c)

## Rationale

- Chemotherapy reduces **recurrence** only; BCG (bacille Calmette-Guérin) reduces **progression** but fails in about a third.
- Combinations add **different mechanisms**; useful in **BCG shortage** and **BCG-unresponsive** NMIBC (non-muscle-invasive bladder cancer).

## 1. Sequential chemotherapy (doublets)

| Regimen | Method | Results |
|---|---|---|
| **Gem/Doce** (gemcitabine–docetaxel) | Gemcitabine 1 g in 50 mL (60–90 min) → drain → docetaxel 37.5 mg (60–120 min); weekly ×6, then monthly for 2 years | High-grade recurrence-free survival about 65% at 1 year, 52% at 2 years; CIS (carcinoma in situ) complete response 60% at 1 year; **84% avoid cystectomy** |
| Gem/MMC (mitomycin C) | Gemcitabine 1 g → MMC 40 mg | Durable recurrence-free about 38% |

- **Why it works:** gemcitabine (antimetabolite) and docetaxel (microtubule stabiliser) act on different cell-cycle phases.
- Oral bicarbonate and antiemetics aid tolerance. No level 1 evidence; phase III BRIDGE compares it with BCG.

## 2. Chemotherapy with BCG

- Sequential MMC–BCG: **no benefit** over BCG alone.
- **Exception: BCG alternating with EMDA (electromotive drug administration)-MMC** in T1: lower recurrence and progression (a later trial non-significant).
- Chemohyperthermia (RITE, radiofrequency-induced thermochemotherapeutic effect) potentiates BCG.

## 3. BCG with immunomodulators

- **BCG + interferon-α2b:** no gain over BCG, but allows BCG dose reduction to one-third.
- **N-803 (nogapendekin alfa)**, an IL-15 (interleukin-15) superagonist activating CD8 T and NK (natural killer) cells. With BCG in BCG-unresponsive CIS: complete response 71%, 45% at 12 months; FDA (Food and Drug Administration)-approved in 2024 (chapter: "under review").

## 4. BCG with checkpoint inhibitors (BCG-naive high risk)

- **CREST** (sasanlimab + BCG) and **POTOMAC** (durvalumab + BCG): improved event-free or disease-free survival.
- **ALBAN** (atezolizumab + BCG): negative. KEYNOTE-676 (pembrolizumab) ongoing.

## 5. Novel combinations

- **Cretostimogene** (oncolytic adenovirus) + pembrolizumab: about two-thirds complete response at 12 months.
- **TAR-200** (gemcitabine-releasing device) ± cetrelimab: very high early complete response in BCG-unresponsive CIS (SunRISe-1).

## Toxicity

- Additive cystitis; immune-related events.

**Diagram to draw:** Gem/Doce sequence timeline (instil, dwell, drain, instil).

**Key points to remember:**

- Gem/Doce: gemcitabine 1 g then docetaxel 37.5 mg; 84% cystectomy-free.
- Only EMDA-MMC improves BCG sequentially.
- N-803 + BCG: CIS complete response 71%.
- BCG + checkpoint: CREST, POTOMAC positive; ALBAN negative.

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## Q3d. Etiology and management of hypospadias fistula.

**Type:** SAQ (10 marks) (also answered in 2015-Summer Paper 4, Q3a)

### Definition and incidence

- A UCF (urethrocutaneous fistula) is a urethral leak anywhere below the meatus: the **commonest complication** of hypospadias repair, just under 10% at short-term follow-up.
- After TIP (tubularised incised plate): 5.7% (**Pfistermuller 2015**); proximal repair 3–45%. Usually at the original meatus or coronal margin.

### Aetiology

- **Distal obstruction** (meatal stenosis, stricture): raises voiding pressure and turbulence; the **key correctable cause**.
- **Ischaemia**, oedema, infection, haematoma, tension.
- **Technical:** overlapping suture lines, poorly inverted epithelium, slowly absorbed sutures, **no barrier layer**.
- **Patient:** proximal meatus, glans <14 mm, redo surgery. A long, rigid neourethra raises resistance (Poiseuille's law).

### Prevention

- **Vascularised barrier** (dartos; **tunica vaginalis** in proximal or redo), offset suture lines, stent in complex repairs.

### Evaluation

- Multiple streams, usually within 2 years.
- Under anaesthesia: **bougie calibration** for distal narrowing, **urethroscopy** if irregular, and a **stress test** (compress proximally and distally, inject saline or methylene blue) for hidden tracts.

### Management

- Rarely closes spontaneously. **Wait 6–12 months**; **correct distal obstruction at the same sitting**.

| Fistula | Repair |
|---|---|
| Small, shaft | Excise the epithelialised tract, inverting subepithelial closure with fine absorbable suture, multiple offset layers, **dartos barrier** |
| Large or thin tissue | Trap-door or island skin flap plus second layer |
| Coronal margin | Implies distal narrowing or poor tissue: **redo urethroplasty and glansplasty** |
| With diverticulum | Excise, or reuse the redundant wall as a flap |

- Recurrence about **20%**; reduce with interposed vascularised layers (dartos, tunica vaginalis, Cecil modification when tissue is scarce).
- Catheter: none for simple repairs, 7–14 days for complex.
- **Also know:** preputial fistula does not involve the urethra; follow through puberty.

```
UCF → wait 6–12 months → calibrate, urethroscopy, stress test
 ├─ distal obstruction → correct at same sitting
 ├─ small → excise + layered closure + barrier flap
 └─ large/coronal → flap repair or redo urethroplasty
```

**Diagram to draw:** neourethra, dartos barrier and skin with offset suture lines.

**Key points to remember:**

- Commonest complication, just under 10%; TIP 5.7%.
- Exclude distal obstruction; wait 6–12 months.
- Layered closure with barrier flap; recurrence about 20%.

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## Q3e. Tubeless PCNL

**Type:** SAQ (10 marks) (also answered in 2016-Summer Paper 4, Q3e and 2025-Winter Paper 2, Q3e)

### Definition

- PCNL (percutaneous nephrolithotomy) that ends **without a nephrostomy tube**.
- **Tubeless**: drainage by a ureteral stent (double-J, or externalised stent or string). **Totally tubeless**: no stent and no tube.
- **Wickham (1984)** did the first tubeless PCNL; **Bellman (1997)** popularised it.

### Rationale

- A large nephrostomy tube causes pain, more analgesia, longer stay and higher cost. Most tracts stop bleeding without tamponade.

### Selection

| Suitable | Keep a nephrostomy |
|---|---|
| Complete clearance, single tract | Pyonephrosis or infected stone |
| Minimal bleeding | Significant bleeding |
| Intact collecting system | Significant collecting-system injury |
| Supracostal access, bilateral cases (shown safe) | Residual stone needing a second look |

### Technique (exit strategy)

1. Confirm clearance by rigid and **flexible nephroscopy** and fluoroscopy.
2. Antegrade double-J stent, or externalised stent or string.
3. Withdraw the sheath under vision; the **safety wire is removed last**.
4. Watch the tract and apply pressure (rib as backstop in thin patients). Fibrin glue did not reduce bleeding in randomised trials.
5. Urethral catheter overnight.

### Evidence

- Meta-analyses of randomised trials: shorter stay and operative time.
- Less pain than large tubes (**>16 Fr**), but **no benefit over small 8–9 Fr tubes**.
- No difference in fever, infection, transfusion or stone-free rate.
- Stent discomfort: worse WiSQoL (Wisconsin Stone Quality of Life) scores than nephrostomy in two trials. Externalised stents come out before discharge without cystoscopy.
- Children: insufficient evidence.

### Disadvantages and complications

- Loss of tract for a second look.
- Stent symptoms and a further procedure for removal.
- Urine leak or urinoma.
- Bleeding without tamponade: re-insert a balloon or Kaye catheter, then **angioembolisation**.

### Current position

- Choose tubeless, small tube (≤10 Fr) or stent case by case; both stents and small tubes are acceptable (AUA, American Urological Association). Pairs well with mini-PCNL.

**Diagram to draw:** kidney with sealed tract and double-J stent.

**Key points to remember:**

- Tubeless = stent only; totally tubeless = neither.
- Benefit is mainly over large (>16 Fr) tubes.
- Keep a tube for bleeding, perforation, infection or second look.

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## Q3f. Miniperc

**Type:** SAQ (10 marks) (also answered in 2017-Winter Paper 4, Q3d and 2022-Summer Paper 1, Q1)

### Definition

- PCNL (percutaneous nephrolithotomy) through a **smaller tract**, to reduce bleeding and renal trauma. First "mini-perc": **Jackman (1998)**, 13 Fr, in children.

| Designation | Tract size |
|---|---|
| Standard | 24–34 Fr |
| **Mini** | 15–22 Fr |
| Ultra-mini | 11–14 Fr |
| Micro-perc | <10 Fr (single-step "all-seeing needle") |

- The nomenclature is contested. The **working channel**, not sheath size, decides which instruments pass.

### Instruments and technique

- Mini nephroscopes (e.g. 12 Fr telescope, 15–18 Fr sheath, 6 Fr channel); holmium or thulium laser, or small pneumatic probes.
- **Fragment removal:** irrigation outflow around the scope washes fragments out (**"vacuum-cleaner" effect**), or **active suction sheaths** (super-mini PCNL).
- Prone or supine; ECIRS (endoscopic combined intrarenal surgery) possible.
- Exit: often **tubeless** or a small nephrostomy (≤10 Fr: less pain).

### Advantages over standard PCNL

- Less haemoglobin drop, **fewer transfusions**, shorter stay; less parenchymal damage.
- Children: a 24–30 Fr tract in a child equals about a 72 Fr tract in an adult, so miniaturisation matters most here.

### Disadvantages

- **Longer operative time**, poorer vision, slower fragment clearance.
- **Intrarenal pressure:** high-pressure irrigation through a tight sheath causes pyelovenous backflow, **fever and sepsis**. Keep outflow free; use suction sheaths.

### Evidence

- Mini versus standard (meta-analysis): **non-inferior stone-free rate**, less bleeding, shorter stay, longer surgery; same adverse events.
- Stones >2 cm: equal stone-free rate and fever; mini bleeds less than **30 Fr** tracts, but **every advantage disappears against 24 Fr**.
- Mini-PCNL versus RIRS (retrograde intrarenal surgery), large matched study: similar stone-free and retreatment rates; RIRS shorter stay.

### Indications and contraindications

- Renal stones **1–2 cm**, **lower-pole stones** >1 cm, failed SWL (shock wave lithotripsy) or RIRS, **paediatric stones**, calyceal diverticular stones; selected larger stones or extra staghorn tracts.
- **Contraindications:** as for PCNL: untreated infection, uncorrected coagulopathy, pregnancy.

**Diagram to draw:** tract sizes compared (standard, mini, ultra-mini, micro).

**Key points to remember:**

- Mini 15–22 Fr; ultra-mini 11–14 Fr; micro <10 Fr; Jackman 1998.
- Fewer transfusions and shorter stay, but longer operation.
- Watch intrarenal pressure; advantage lost against 24 Fr tracts.

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# Rapid Revision Sheet – Practice Set 6 Paper 4

*All the key points from this paper in one place. Read this in the last 30 minutes before the exam.*

**Q1. Present concept in management of hypospadias (LAQ)**

- Classify after degloving (GMS); curvature, not meatus, decides one versus two stages.
- TIP for distal; staged graft repair (STAG/Bracka) for severe proximal.
- Hypospadias plus undescended testis: DSD work-up.
- Follow through puberty; use patient-reported outcomes.

**Q2. Discuss recent advances in tissue engineering with respect to urology (LAQ)**

- Triad: cells, scaffold, signals; vascularity (100–200 µm diffusion) is the main barrier.
- Atala 2006 promising; phase II trial no gain; conduit trial half stenosed.
- Raya-Rivera 2011: seeded urethras in 5 boys.
- New: iPSCs, urine-derived cells, bioprinting, organoids.

**Q3a. Newer methods of stone analysis and their significance in management of calculus disease. (SAQ)**

- FTIR and XRD are reference methods; wet chemistry is obsolete.
- DECT separates uric acid stones in vivo.
- Composition guides dissolution, SWL choice and prevention.

**Q3b. Urolift. (SAQ)**

- 30–80 mL, no median lobe; local anaesthesia.
- Ejaculation preserved; retreatment 13.6% at 5 years.
- TURP better on IPSS, flow and durability.

**Q3c. Combination Intravesical therapy (SAQ)**

- Gem/Doce: gemcitabine 1 g then docetaxel 37.5 mg; 84% cystectomy-free.
- Only EMDA-MMC improves BCG sequentially.
- N-803 + BCG: CIS complete response 71%.
- BCG + checkpoint: CREST, POTOMAC positive; ALBAN negative.

**Q3d. Etiology and management of hypospadias fistula. (SAQ)**

- Commonest complication, just under 10%; TIP 5.7%.
- Exclude distal obstruction; wait 6–12 months.
- Layered closure with barrier flap; recurrence about 20%.

**Q3e. Tubeless PCNL (SAQ)**

- Tubeless = stent only; totally tubeless = neither.
- Benefit is mainly over large (>16 Fr) tubes.
- Keep a tube for bleeding, perforation, infection or second look.

**Q3f. Miniperc (SAQ)**

- Mini 15–22 Fr; ultra-mini 11–14 Fr; micro <10 Fr; Jackman 1998.
- Fewer transfusions and shorter stay, but longer operation.
- Watch intrarenal pressure; advantage lost against 24 Fr tracts.